

Library Management System Requirements Specification

Project Goal

Build an ASP.NET Core Web API project for learning CRUD operations and core ASP.NET concepts using 3-Layer Architecture without Authentication initially, while keeping the project ready for JWT integration later.

Architecture

- Library.API (Presentation Layer)
- Library.BLL (Business Logic Layer)
- Library.DAL (Data Access Layer)
- Optional: Library.Domain for shared entities

Entities

Author

- Id
- FullName
- Biography
- DateOfBirth
- Nationality

Category

- Id
- Name
- Description

Publisher

- Id
- Name

- Address
- Phone

Book

- Id
- Title
- ISBN
- PublicationYear
- TotalCopies
- AvailableCopies
- Price
- AuthorId
- CategoryId
- PublisherId

Member

- Id
- FullName
- Email
- Phone
- Address
- RegistrationDate

Borrowing

- Id
- BookId
- MemberId
- BorrowDate
- DueDate
- ReturnDate

- Status

Relationships

- Author 1..* Books
- Category 1..* Books
- Publisher 1..* Books
- Member 1..* Borrowings
- Book 1..* Borrowings

Required Endpoints

Authors

GET, GET by Id, POST, PUT, DELETE

Categories

GET, GET by Id, POST, PUT, DELETE

Publishers

GET, GET by Id, POST, PUT, DELETE

Books

GET, GET by Id, POST, PUT, DELETE

Search, Filter, Pagination, Sorting

Members

GET, GET by Id, POST, PUT, DELETE

Borrowings

Borrow book, Return book, Get overdue books, Get member borrowings.

Business Rules

- A book cannot be borrowed if AvailableCopies = 0.
- Borrow duration is 14 days.
- Author deletion is not allowed when books exist.
- AvailableCopies decreases on borrow and increases on return.

DTO Usage

- Create DTO
- Update DTO
- Response DTO
- Details DTO

ASP.NET Concepts Covered

- Controllers
- Routing
- Dependency Injection
- Entity Framework Core
- DTOs
- Repository Pattern
- Service Layer
- Validation
- Middleware
- Exception Handling
- Swagger
- Logging
- Pagination
- Filtering
- Sorting
- Searching

Future Enhancements

- JWT Authentication
- Authorization
- Roles
- Refresh Tokens
- Soft Delete
- Caching
- API Versioning